

Synoptic CB: Macrophyte Fluxes

2025 Samples

2026-07-27

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#Read in and collate all the files in the Processed Data folder

##Pull in Hydrago probe data

Some flux IDs are missing from hydrago data

```
##      missing_ids
## 1  GCW_TR_1_202509
## 2  GCW_TR_2_202509
## 3  GCW_TR_3_202509
## 4  GCW_TR_4_202509
## 5  GCW_TR_5_202509
## 6  GCW_WC_1_202509
## 7  GCW_WC_2_202509
## 8  GCW_WC_3_202509
## 9  GCW_WC_4_202509
## 10 GCW_WC_5_202509
## 11 GWI_WC_1_202507
## 12 GWI_WC_2_202507
## 13 GWI_WC_3_202507
## 14 GWI_WC_4_202507
## 15 GWI_WC_5_202507
```

##pull TEROS data for the zone during the time of the fluxes and put it into another column so that we can use that too.

0.1 Match up the TEROS soil data and add it to the flux dataframe

```
## Remove bad fluxes and calculate how many there were
```

```
## Number flux measurements removed: 15
```

```
## Number negative CO2 fluxes removed: 1
```

```
## Number unusually high starting value fluxes removed: 3
```

```
## Number messy fluxes removed: 11
```

```
## Percentage flux measurements removed: 8.82%
```

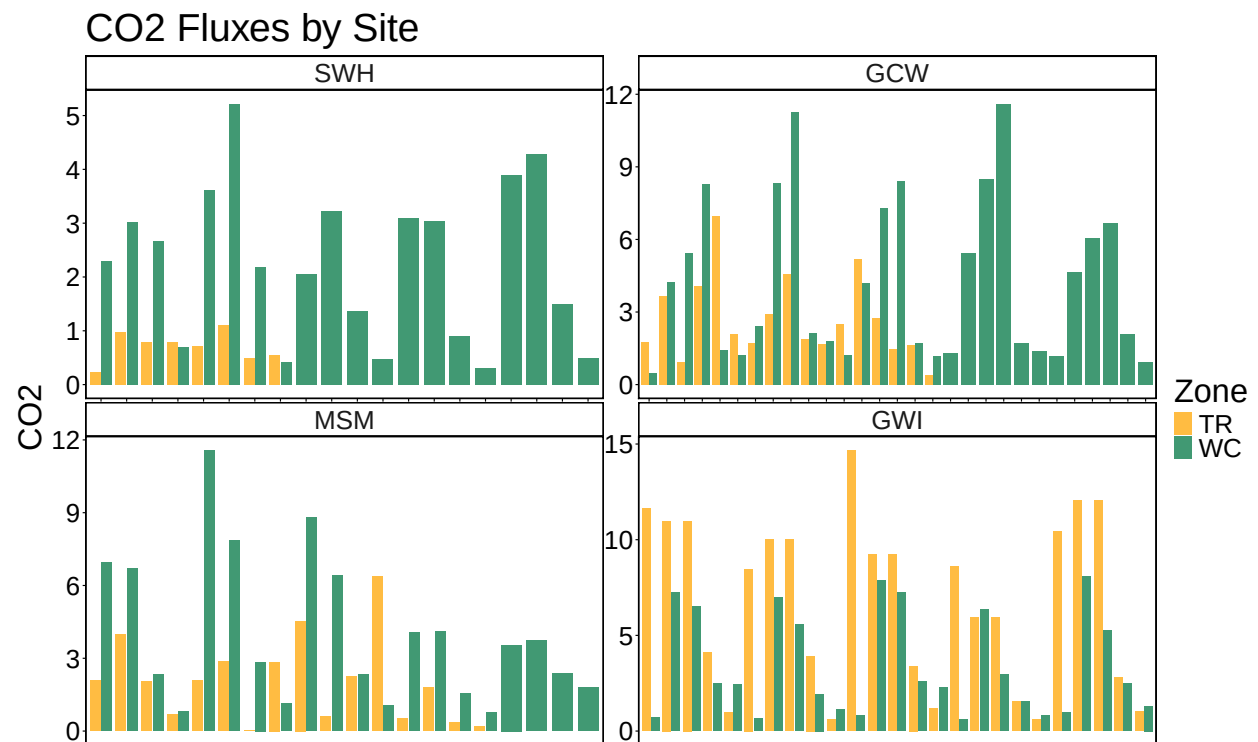
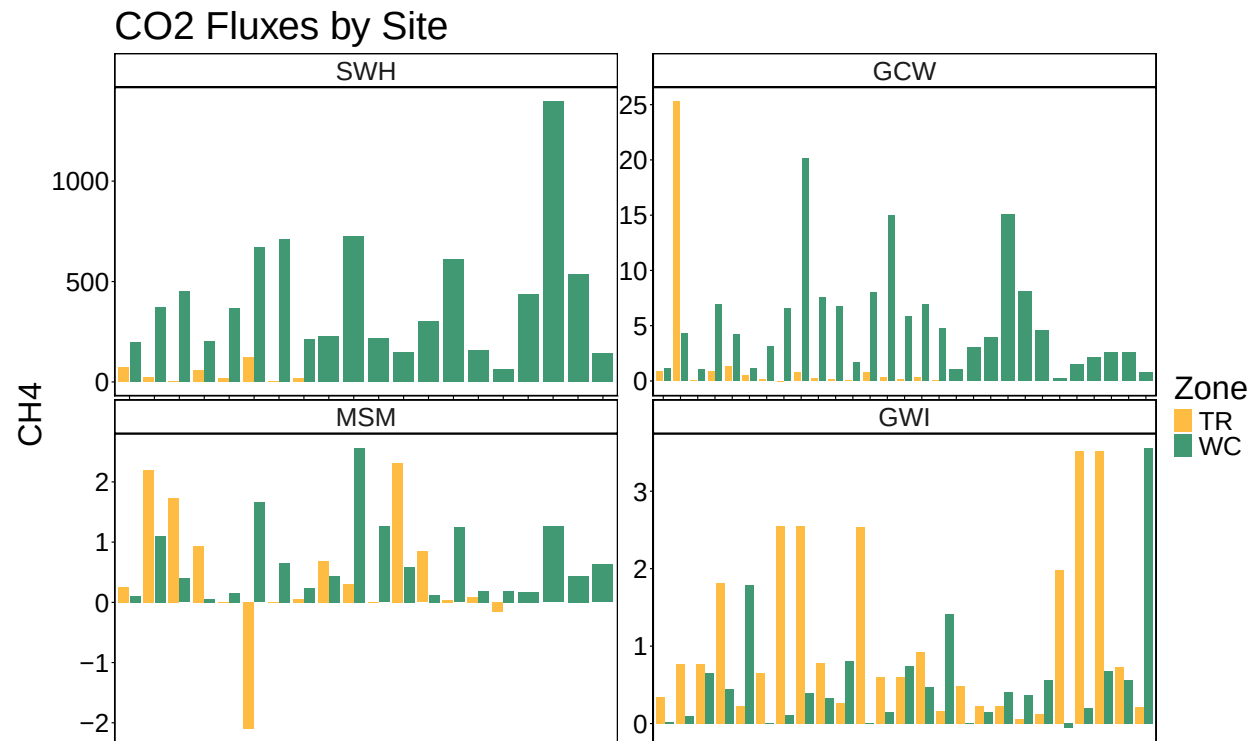
```
## Percentage negative CO2 fluxes removed: 0.59%
```

```
## Percentage unusually high starting value fluxes removed: 1.76%
```

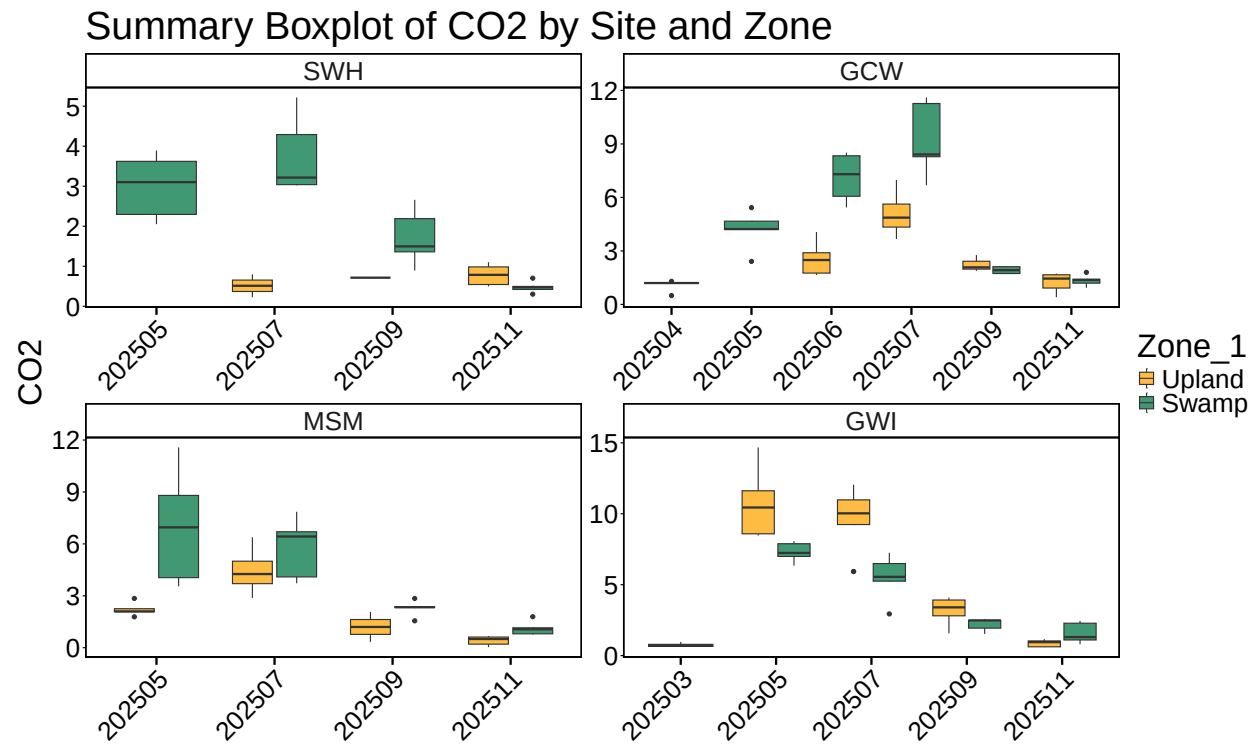
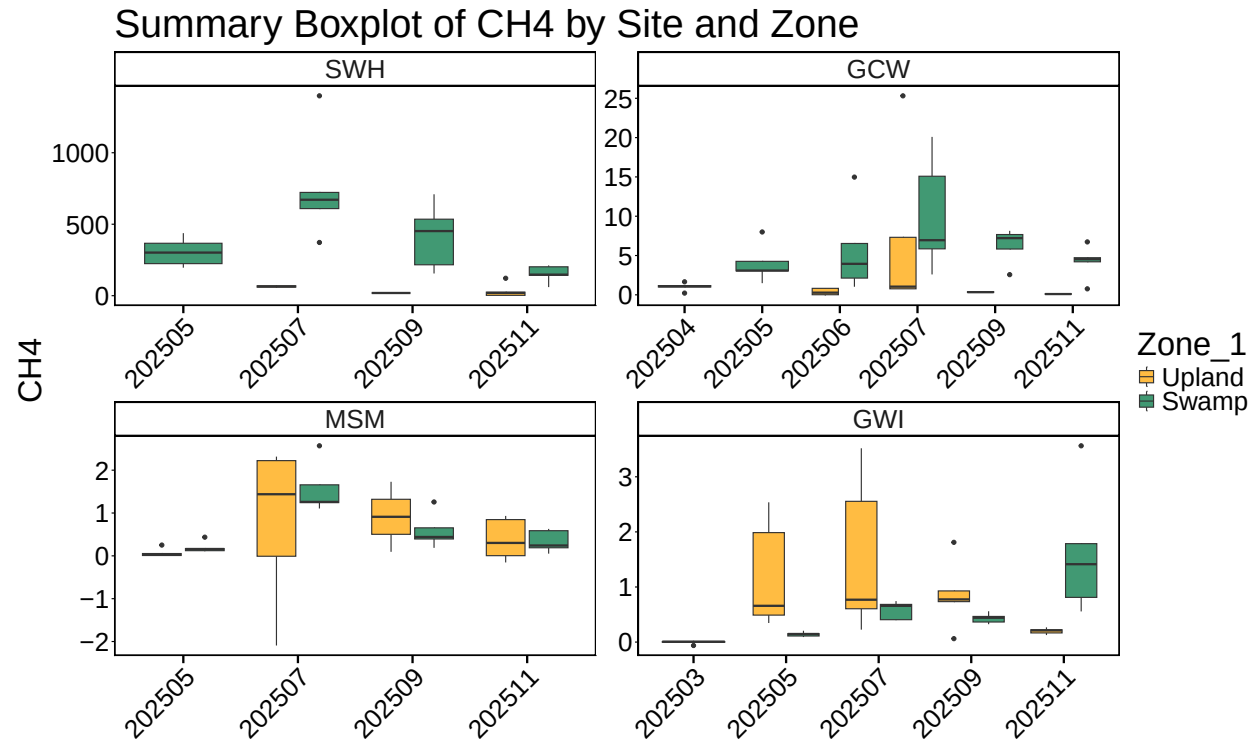
```
## Percentage messy fluxes removed: 6.47%
```

0.2 Write out files

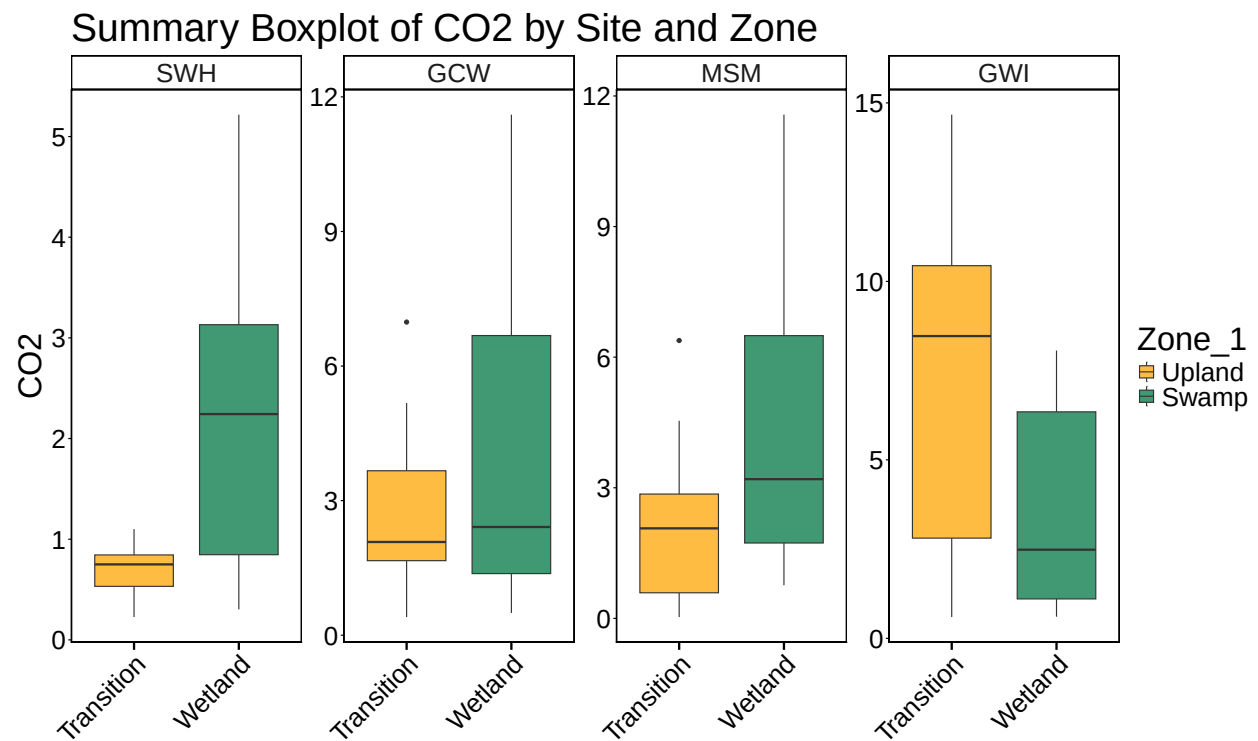
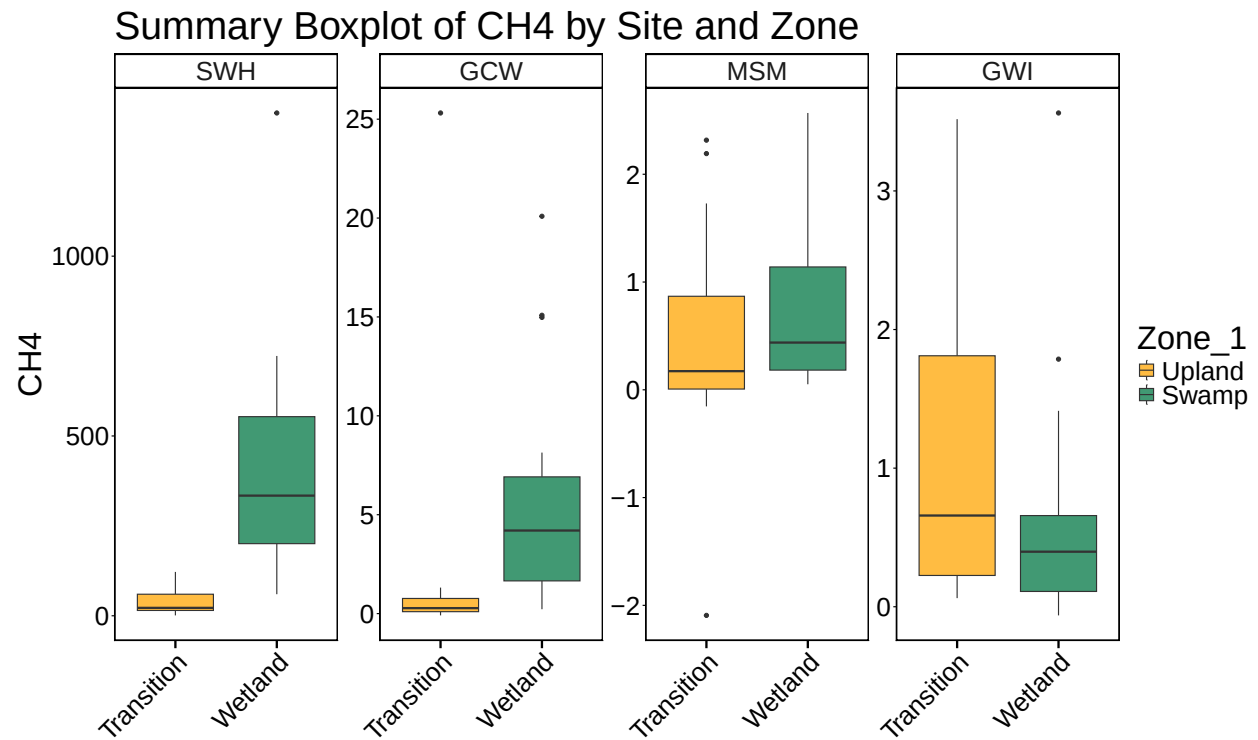
0.3 All Data by Year, Site and Zone



0.4 Boxplot summary by Date



0.5 Boxplot summary by Site



##Calculate summary metrics

0.6 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Transition	1.0872692	121.977500	39.7087775	0.2266990	1.100404	0.7058451
SWH	Wetland	59.9871778	1398.120534	406.5684788	0.3028537	5.217695	2.2376064
GCW	Transition	-0.0901019	25.312906	1.8577613	0.4050037	6.980726	2.7131738
GCW	Wetland	0.2215926	20.090639	5.1816997	0.4957703	11.604588	4.2294688
MSM	Transition	-2.0931613	2.316486	0.4504229	0.0305177	6.382414	2.0786885
MSM	Wetland	0.0514141	2.568636	0.6728730	0.7602931	11.571173	4.0361589
GWI	Transition	0.0621000	3.517843	1.0666977	0.5968420	14.671811	6.8158948
GWI	Wetland	-0.0623885	3.562404	0.5542746	0.6065226	8.062884	3.4680774